PRAKTIKUM BASIS DATA

MODUL 8

SCRIPTING SQL (CREATE INDEX, CREATE VIEW, ANALYZE TABLE)

Disusun Oleh:

Ghina Hasna Putri Tinimada (103122400031)

PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

2026

1. **MEMBUAT DESAIN DATABASE UNTUK SUATU PERUSAHAAN :**

   a. Table department dengan primary key adalah department_id. Tabel departemen terdiri atas field-field: department_id, department_name, manager_id, dan location_id.

   Query :

```
CREATE TABLE department (
    department_id INT PRIMARY KEY,
    department_name VARCHAR(100) NOT NULL,
    manager_id INT,
    location_id INT
);
```

   Output :

Columns | Data | Model | Constraints | Grants | Statistics | Triggers | Flashback | Dependencies | Details | Partitions | Indexes | SQ

Actions...

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	DEPARTMENT_ID	NUMBER(38,0)	No	(null)	1	(null)
2	DEPARTMENT_NAME	VARCHAR2(100 BYTE)	No	(null)	2	(null)
3	MANAGER_ID	NUMBER(38,0)	Yes	(null)	3	(null)
4	LOCATION_ID	NUMBER(38,0)	Yes	(null)	4	(null)

   b. Table employee dengan constraint foreign key adalah department_id. Tabel employee terdiri atas field-field: employee_id, last_nama NOT NULL, email, salary, commission_pct, hire_date NOT NULL

   Query :

```
CREATE TABLE employee (
    employee_id INT PRIMARY KEY,
    first_name VARCHAR(50),
    last_name VARCHAR(50) NOT NULL,
    email VARCHAR(100),
    salary DECIMAL(10,2),
    commission_pct DECIMAL(4,2),
    hire_date DATE NOT NULL,
    department_id INT,

    CONSTRAINT fk_department
    FOREIGN KEY (department_id)
    REFERENCES department(department_id)
);
```

   Output :

```
Table EMPLOYEE created.
```

c. Tambahkan tabel-tabel lain yang saling berelasi (seperti: tabel persediaan barang, pemasok barang, dst).

- Tabel Supplier

  Query :

```sql
CREATE TABLE supplier (
    supplier_id INT PRIMARY KEY,
    supplier_name VARCHAR(100) NOT NULL,
    phone VARCHAR(20),
    address VARCHAR(150)
);
```

  Output :

```
Table SUPPLIER created.
```

- Tabel Product

  Query :

```sql
CREATE TABLE product (
    product_id INT PRIMARY KEY,
    product_name VARCHAR(100) NOT NULL,
    stock INT,
    price DECIMAL(10,2),
    supplier_id INT,

    CONSTRAINT fk_supplier
    FOREIGN KEY (supplier_id)
    REFERENCES supplier(supplier_id)
);
```

  Output :

```
Table PRODUCT created.
```

2. ANALISIS TABLE-TABLE DENGAN COMPUTE STATISTICS

Query :

```sql
ANALYZE TABLE department COMPUTE STATISTICS;

ANALYZE TABLE employee COMPUTE STATISTICS;

ANALYZE TABLE supplier COMPUTE STATISTICS;

ANALYZE TABLE product COMPUTE STATISTICS;
```

Output :

```
Table DEPARTMENT analyzed.


Table EMPLOYEE analyzed.


Table SUPPLIER analyzed.


Table PRODUCT analyzed.
```

3. ISIKAN DATA-DATA YANG BERSESUAIAN DENGAN TABEL-TABEL PADA NOMOR 1 (MASING-MASING 10 RECORD).

a. Department

Worksheet Query Builder

```sql
INSERT ALL
    INTO department VALUES (1, 'Human Resource', 101, 10)
    INTO department VALUES (2, 'Finance', 102, 20)
    INTO department VALUES (3, 'IT Support', 103, 30)
    INTO department VALUES (4, 'Marketing', 104, 40)
    INTO department VALUES (5, 'Production', 105, 50)
    INTO department VALUES (6, 'Sales', 106, 60)
    INTO department VALUES (7, 'Logistic', 107, 70)
    INTO department VALUES (8, 'Customer Service', 108, 80)
    INTO department VALUES (9, 'Research', 109, 90)
    INTO department VALUES (10, 'Security', 110, 100)
SELECT * FROM dual;
```

Output :

	DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
1	1	Human Resource	101	10
2	2	Finance	102	20
3	3	IT Support	103	30
4	4	Marketing	104	40
5	5	Production	105	50
6	6	Sales	106	60
7	7	Logistic	107	70
8	8	Customer Service	108	80
9	9	Research	109	90
10	10	Security	110	100

b. Employee

Query :

```sql
INSERT ALL
    INTO employee VALUES (201, 'Ghina', 'Hasna', 'ghina1@gmail.com', 5000000, 0.10, DATE '2023-01-10', 1)
    INTO employee VALUES (202, 'Ghina', 'Hasna', 'ghina2@gmail.com', 5500000, 0.15, DATE '2023-02-15', 2)
    INTO employee VALUES (203, 'Ghina', 'Hasna', 'ghina3@gmail.com', 6000000, 0.12, DATE '2023-03-20', 3)
    INTO employee VALUES (204, 'Ghina', 'Hasna', 'ghina4@gmail.com', 6500000, 0.11, DATE '2023-04-11', 4)
    INTO employee VALUES (205, 'Ghina', 'Hasna', 'ghina5@gmail.com', 7000000, 0.09, DATE '2023-05-05', 5)
    INTO employee VALUES (206, 'Ghina', 'Hasna', 'ghina6@gmail.com', 5200000, 0.08, DATE '2023-06-17', 6)
    INTO employee VALUES (207, 'Ghina', 'Hasna', 'ghina7@gmail.com', 5800000, 0.10, DATE '2023-07-19', 7)
    INTO employee VALUES (208, 'Ghina', 'Hasna', 'ghina8@gmail.com', 6200000, 0.13, DATE '2023-08-21', 8)
    INTO employee VALUES (209, 'Ghina', 'Hasna', 'ghina9@gmail.com', 6700000, 0.14, DATE '2023-09-12', 9)
    INTO employee VALUES (210, 'Ghina', 'Hasna', 'ghina10@gmail.com', 7200000, 0.16, DATE '2023-10-01', 10)
SELECT * FROM dual;

COMMIT;
```

Output :

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	SALARY	COMMISSION_PCT	HIRE_DATE	DEPARTMENT_ID
1	201	Ghina	Hasna	ghina1@gmail.com	5000000	0,1	10-01-2023	1
2	202	Ghina	Hasna	ghina2@gmail.com	5500000	0,15	15-02-2023	2
3	203	Ghina	Hasna	ghina3@gmail.com	6000000	0,12	20-03-2023	3
4	204	Ghina	Hasna	ghina4@gmail.com	6500000	0,11	11-04-2023	4
5	205	Ghina	Hasna	ghina5@gmail.com	7000000	0,09	05-05-2023	5
6	206	Ghina	Hasna	ghina6@gmail.com	5200000	0,08	17-06-2023	6
7	207	Ghina	Hasna	ghina7@gmail.com	5800000	0,1	19-07-2023	7
8	208	Ghina	Hasna	ghina8@gmail.com	6200000	0,13	21-08-2023	8
9	209	Ghina	Hasna	ghina9@gmail.com	6700000	0,14	12-09-2023	9
10	210	Ghina	Hasna	ghina10@gmail.com	7200000	0,16	01-10-2023	10

4. ANALISIS TABLE-TABLE DENGAN COMPUTE STATISTICS

Query :

```
ANALYZE TABLE department COMPUTE STATISTICS;

ANALYZE TABLE employee COMPUTE STATISTICS;

ANALYZE TABLE supplier COMPUTE STATISTICS;

ANALYZE TABLE product COMPUTE STATISTICS;
```

Output :

```
Table DEPARTMENT analyzed.


Table EMPLOYEE analyzed.


Table SUPPLIER analyzed.


Table PRODUCT analyzed.
```

5. BUAT VIEW EMPVU80 YANG BERISI ID_NUMBER, NAME, SALARY, DEPARTMENT_ID DARI PEGAWAI YANG BEKERJA DI DEPARTMENT = 80.

Query :

```
CREATE VIEW empvu80 AS
SELECT
    employee_id AS id_number,
    first_name || ' ' || last_name AS name,
    salary,
    department_id
FROM employee
WHERE department_id = 8;
```

Output :

```
View EMPVU80 created.
```

6. **KEMUDIAN TAMPILKAN STRUKTUR DARI VIEW EMPVU80.**

Query :

```
DESCRIBE empvu80;
```

Output :

```
Name          Null?    Type
------------- -------- -------------
ID_NUMBER     NOT NULL NUMBER(38)
NAME                   VARCHAR2(101)
SALARY                 NUMBER(10,2)
DEPARTMENT_ID          NUMBER(38)
```

7. **BUAT NON-UNIQUE INDEX PADA KOLOM FOREIGN KEY YANG ADA PADA TABEL EMPLOYEE.**

Query :

```
CREATE INDEX idx_department
ON employee(department_id);
```

Output :

```
Index IDX_DEPARTMENT created.
```

```
Index IDX_DEPT_ID created.
```